

Exact Analytic Orbits of a Nonlinear p -Adic Double Shear: Joint Interpolation on the Original Clock

HCS-C394 theorem and reproducibility package

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Abstract

A nonlinear symplectic double shear on the p -adic unit polydisc admits an analytic extension of its original integer iteration clock. We give the joint Tate-algebra construction at every nonzero parameter with valuation at least one for odd primes and at least two at the prime two. The origin is its only genuine periodic point. Independent exact computations audit the formulas without replacing their proofs. The interpolation and analytic-zero methods have classical owners; no new certificate algorithm or target prime-period identity is claimed.

Keywords: p -adic dynamics; symplectic shear; analytic interpolation; minimal component; residue period; algebraic return.

中文摘要

本文研究单位多圆盘上的非线性辛双剪切映射，将原始整数迭代时间联合解析地延拓到素进整数。奇素数下参数估值至少为一，二进情形至少为二，参数零单独排除。原点是完整素进空间内唯一真正的周期点。精确有限计算只检验公式的实现。经典插值理论明确归属，不将原生算术结构写成目标零点对应。关键词：素进动力学；辛剪切；解析插值；最小分量；剩余周期；代数返回。

1 One nonlinear source and its arithmetic boundary

Let p be any rational prime and normalize $v_p(p) = 1$. Fix a nonzero $a \in \mathbb{Z}_p$ with $c = v_p(a) \geq 1$ when p is odd and $c \geq 2$ when $p = 2$. On $X = \mathbb{Z}_p^2$ use the maximum norm and freeze

$$F_a(x, y) = (x + ay^2, y + a(x + ay^2)^2). \quad (1)$$

The coefficientwise proximity to the identity permits analytic time interpolation, but proximity alone does not describe all orbits. For this map, the additional homogeneous shear structure determines the exact radius-dependent clock. Finite residue permutations and periodic points in X must remain separate throughout.

Poonen's joint interpolation theorem and its dyadic boundary [1] supply the classical method reconstructed below. The analytic zero-set approach to dynamical Mordell–Lang belongs to Bell–Ghioca–Tucker [2]. Recent finite-precision and refined-tail certificate methods belong to Mukhamedov [3]; we claim no new effective certification algorithm. Our question is the full orbit structure of the particular nonlinear source, not a new general interpolation theorem.

The nearest project mechanisms are C166's finite affine dyadic Pascal tower, C174's expanding parity renewal, C258's mixed congruential map, and C283's conductor-shell heat operator. None supplies the nonlinear joint analytic action and valuation-scale decomposition of (1). The symbolic branch's P23 imports classical linear-recurrence zero-set rigidity, not this nonlinear analytic-arc proof. These comparisons define the repository increment; they are not a global novelty certification.

2 Symplecticity, reversal, and joint analytic time

Write $F_a = S_y S_x$ for the two shears $S_x(x, y) = (x + ay^2, y)$ and $S_y(x, y) = (x, y + ax^2)$. Both preserve $dx \wedge dy$, and

$$F_a^{-1}(x, y) = (x - a(y - ax^2)^2, y - ax^2). \quad (2)$$

The involution $R_0(x, y) = (-y, -x)$ satisfies $R_0 S_x R_0 = S_y^{-1}$ and $R_0 S_y R_0 = S_x^{-1}$, hence $R_0 F_a R_0 = F_a^{-1}$ on the same clock. Integral coefficients in (1)–(2) make both maps 1-Lipschitz; each is therefore an isometry and induces a permutation modulo every p^N .

Theorem 1. *There is $G_a \in \mathbb{Z}_p\langle x, y, t \rangle^2$ such that $G_a(u, n) = F_a^n(u)$ for every integer $n \geq 0$ and*

$$G_a(G_a(u, s), t) = G_a(u, s + t) \quad (s, t \in \mathbb{Z}_p).$$

Thus negative time is the inverse of positive time, without a new roof.

Proof. On $A = \mathbb{Z}_p\langle x, y \rangle$, set $\Delta h = h \circ F_a - h$. Factoring differences of monomials and taking Gauss-norm limits gives $\|\Delta h\|_p \leq p^{-c} \|h\|_p$. In particular $\Delta^m u \in p^{cm} A^2$. The joint series

$$G_a(u, t) = \sum_{m \geq 0} \binom{t}{m} \Delta^m u \quad (3)$$

converges in the Tate algebra because its m th term has coefficient norm at most $p^{-cm+v_p(m!)}$ and $v_p(m!) \leq m/(p-1)$. If $\delta = c - 1/(p-1) > 0$, its tail after M has norm at most $p^{-(M+1)\delta}$. At integer $n \geq 0$ the series truncates to $(I + \Delta)^n u = F_a^n(u)$. The action law holds first at nonnegative integer pairs. Such pairs are dense in $\mathbb{Z}_p^2$ and all compositions are continuous and integral-valued, so the law holds at every pair. The time-zero map is the identity. $\square$

The sufficient threshold cannot be replaced by a general dyadic coefficient congruence modulo two: $f(x) = -x$ has a nonzero two-cycle. An analytic interpolant of the orbit of one would equal one on all even times and therefore identically, contradicting its odd-time value. This general control does not assert failure of every double shear outside the stated threshold. Pointwise residue identity also does not suffice: $x^p = x$ on $\mathbb{F}_p$, but $x^p - x$ has unit coefficients.

3 Genuine periods and the first revision boundary

The action has no nontrivial finite-time stabilizer at a nonfixed point. Indeed, if $F_a^n(u) = u$ for some $n \geq 1$, then every coordinate of $G_a(u, t) - u$ vanishes at the infinite set $t = kn$. A nonzero restricted one-variable analytic function has finitely many zeros on $\mathbb{Z}_p$ by Strassmann's theorem. Hence the coordinates vanish identically and $F_a(u) = u$. Solving (1) gives $ay^2 = 0$ and $a(x + ay^2)^2 = 0$, whence $u = 0$. At this point $DF_a(0) = I$, so it is parabolic rather than expanding. The genuine point-counting zeta is $(1 - z)^{-1}$.

Round zero establishes the source and joint clock, but does not yet determine nonperiodic orbit closures or all finite quotient periods. Those require the strict displacement estimate developed in the next revision. Neither a finite quotient permutation nor the preceding scalar zeta constructs a target determinant.

Revision record

Round zero: source automorphism, joint analytic time and genuine-period collapse. Exact orbit scales and finite-level classification are deferred.

References

- [1] B. Poonen, *p-adic interpolation of iterates*, Bull. Lond. Math. Soc. **46** (2014), 525–527. doi:10.1112/blms/bdu010.
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